TaurEye — CFO Financial Review

All-in input costs, revenue models, unit economics & break-even, and the margin plan · Indicative figures (verify vendor pricing) · $1 = ₹88

1. Executive summary

TaurEye is an ad-subsidised freemium EOD stock app. The cost base is unusually light because the product is static and compute runs client-side — the only material variable cost is the LLM (AI analysis/reports), and the dominant revenue is advertising, topped up by credit purchases, broker affiliate and a future Pro subscription.

Against a 1M-user base (mix: 10% light / 15% medium / 5% heavy / 70% dormant) the model lands at break-even ≈ 4% paying conversion, and is marginally profitable (≈ +₹1.76L/month) once free reports are gated behind a rewarded ad. The plan below widens that margin and removes the single-stream ad dependency.

CFO verdict: viable at launch; margin improves with scale if we (a) cache AI output, (b) monetise free reports with rewarded ads, (c) switch on broker affiliate now, and (d) diversify off ads.

2. Input cost structure (all tools)

Free at scale (if architected right): Oracle VM (Always-Free), Cloudflare CDN (free), GitHub Actions (public repo), FCM push, NSE/BSE data, self-hosted Capgo OTA.

Item	Type	Trigger	Cost (at scale)
LLM — Anthropic (Haiku)	COGS / usage	per AI report (~₹0.70)	**₹0.35L–₹10.5Cr/mo***
Supabase Pro	Infra / time+usage	always-on, MAU, backups	**$25 base → ~₹3.0–3.7L/mo @1M MAU**
Apple Developer	Distribution / time	annual, iOS publish	**$99/yr (~₹700/mo)**
Google Play	Distribution / one-time	Android publish	**$25 once**
Domain	Infra / time	annual renewal	**~₹100/mo**
SMTP (auth email)	Infra / usage	signup/reset volume	**~₹1,500/mo**
Razorpay	Payments / per-txn	on credit sales	**~2% + 18% GST**
Capgo OTA	Distribution / usage	MAU (or self-host)	**₹0 self-hosted**
Monitoring (optional)	Ops / usage	event volume	**₹0–8k/mo**
Kite Connect (future)	Feature / time	only if live broker sync	**₹500/mo**

*LLM is the swing item — ₹0.35L/mo (light) to ₹10.5Cr/mo (every user 5 reports/day). Caching caps it at ~₹2.4L/mo total, independent of user count (see §5). Fixed infra opex ≈ ₹3.5L/mo, ~95% Supabase.

3. Revenue models

Stream	Mechanism	Rate (conservative)	Role
Banner ads	Free users only	eCPM ~₹12 → ARPU ~₹8.7/free user/mo	~55% of revenue
Rewarded ads	1 ad per free report	eCPM ~₹80 → ₹0.08/report	Monetises free-AI bleed
Interstitial ads	Between screens	eCPM ~₹50	Incremental
Credit packs	Razorpay (Phase C)	₹99→100, ₹299→350, ₹599→750 cr	High-intent paywall
Broker affiliate	Demat signups	₹300–500 / account	Near-100% margin, fastest

Stream	Mechanism	Rate (conservative)	Role
Pro subscription	Phase D	₹999/yr (ad-free + AI quota)	Highest LTV, recurring

Per paid report: charge 5 credits (~₹4.25) vs ~₹0.70 LLM = ~84% margin (→ ~99% with caching). Broker affiliate at 3% of 1M users ≈ ₹1.2 Cr one-time — currently un-wired (money left on table).

4. Unit economics — base case (1M users)

Monthly P&L; (base mix, rewarded-gated)	₹ / month
Report revenue (paid)	**₹12.75 L**
Banner ads (free users)	**₹25.88 L**
Rewarded ads (5M free reports)	**₹4.00 L**
Total revenue	**₹42.63 L**
LLM cost (before caching)	**(₹37.10 L)**
Fixed infra	**(₹3.52 L)**
Payment fees	**(₹0.26 L)**
NET	**+₹1.76 L (+₹21.1 L/yr)**
Break-even paying conversion	**≈ 4.1%**

Without rewarded-gating the same base case is −₹2.0L/mo (break-even ~6%). The rewarded ad on each free report is what flips it positive. Model is configurable in docs/unit-economics.html.

5. Pro-forma — after the cost-down / revenue-up plan

Same 1M-user base, with the plan implemented: AI output cached (LLM decouples from scale), broker affiliate live, Pro at 1% penetration (₹999/yr, ad-free), and ad mediation lifting eCPMs.

Monthly P&L; line	Base case	After plan
Report revenue (paid)	₹12.75 L	**₹12.75 L**
Banner ads (free users)	₹25.88 L	**₹31.63 L**
Rewarded ads	₹4.00 L	**₹5.00 L**
Interstitial ads	—	**₹3.00 L**
Broker affiliate	—	**₹5.00 L**
Pro subscriptions (1%)	—	**₹8.33 L**
TOTAL REVENUE	**₹42.63 L**	**₹65.70 L**
LLM cost	(₹37.10 L)	**(₹2.40 L) cached**
Fixed infra	(₹3.52 L)	**(₹3.50 L)**
Payment fees	(₹0.26 L)	**(₹0.42 L)**
NET / month	**+₹1.76 L**	**+₹59.4 L**
NET / year	**+₹21.1 L**	**+₹7.1 Cr**

Swing drivers: AI caching cuts LLM ₹37.1L → ₹2.4L (+₹34.7L to net); affiliate + Pro + ad uplift add ~₹23L revenue. Gross margin rises from ~13% to ~90% as the only real COGS collapses. Pro-forma — assumes the plan executes (caching shipped, 3% affiliate take-up over time, 1% Pro, eCPM uplift).

6. Cost minimisation plan (ranked by impact)

- Cache AI output per stock, per day (₹0 to build, ~80% LLM cut). Stock reports are factual + EOD — identical for all users that day. Generate once, serve to all, still charge a credit. LLM cost decouples from user count → ceiling ~₹2.4L/mo total; margin 84% → ~99%. Pre-generate the top ~500 stocks in the nightly pipeline = ₹0 marginal.
- Stop giving away the costly item. Rewarded-ad-gate free reports; cut free allowance 30 → 5/mo; heavy AI = paid-only.

- Hold fixed infra flat & cheap. Static + Oracle Always-Free + Cloudflare free + Supabase Pro ($25, not Team) + self-host Capgo. Prune auth audit logs + old ledger rows (pg_cron) to stay on cheap DB tiers far longer.
- Defer paid tools (Kite ₹500/mo, paid data feeds, smallcase) until revenue demands them.

7. Revenue maximisation plan (ranked by ROI)

- Broker affiliate — today. Paste Angel/Upstox/Dhan URLs. ₹300–500/signup; ~₹1.2 Cr one-time at 3% of 1M. Pure margin.
- Optimise the ad stack. Banner (free) + rewarded-per-report + interstitial; AdMob mediation + a 2nd network to lift eCPM 20–40%.
- Credit packs (Razorpay). Paywall at the high-intent moment (free reports exhausted → 'unlock full report'). 4% conversion funds the rest.
- Pro subscription (₹999/yr). Ad-free + AI quota + the institutional portfolio analytics already built — recurring, highest LTV.
- Diversify: drive affiliate + conversion + Pro so no single stream > 50% of revenue (today it's dangerously ad-heavy).

8. 90-day sequence

Phase	Actions	Cost	Impact
0 — now	Broker links · rewarded-gate AI · free=5 · AI caching · prune logs	**₹0**	Flips to profit; caps LLM; +₹1.2Cr affiliate ceiling
1 — launch	HTTPS+Cloudflare · Supabase Pro · Razorpay · ad mediation	**~₹3–4k/mo**	Unlocks paid revenue + reliability
2 — scale	Pro subscription · conversion paywalls · 2nd ad network	**dev time**	Recurring LTV; de-risk ads

9. KPIs & guardrails

Track monthly: gross margin/report · LLM ₹/active user · paying conversion % · ad ARPU (free user) · revenue concentration (no stream >50%) · CAC vs LTV (target ≥3:1) · Supabase MAU · cash runway. Hard guardrail: never let credit price fall below LLM cost (the model flags this).

10. Key risks

- Ad concentration + India's low/volatile eCPM — one eCPM crash and the base case goes negative.
- Conversion is unproven — 0 paying users today; the whole plan rides on reaching ~4%.
- LLM price moves — mitigated heavily by caching.
- Single-VM / Oracle reclamation holding user money; and SEBI / GST exposure as revenue starts.

Appendix A — ₹0 Bootstrap Mode

Truly ₹0 is achievable on cash — paid for in founder time, feature/quality compromises, and scaling ceilings. The method: max free tiers, self-host the rest on the Always-Free VM, and kill the only real COGS (the LLM). Use this to launch & validate, then graduate to the paid stack once revenue covers it.

Cost today	₹0 replacement	Trade-off
LLM (Anthropic)	Templated/rule-based analysis from existing metrics & patterns; or a free-tier LLM (Gemini / Groq / Cloudflare Workers AI) with caps	No frontier AI; free LLM is rate-limited
Supabase Pro ($25)	Free tier + a free daily cron pinger to dodge the 7-day pause	500MB / 50k MAU cap; no backups
Domain + TLS	Cloudflare Pages → free *.pages.dev + HTTPS + CDN	Unbranded URL (also fixes HTTP + single-VM SPOF)
VM serving	VM runs pipeline only; bundle → Cloudflare R2 free (10GB, zero egress)	Static-only on the free host
Apple $99 + Play $25	PWA (manifest already shipped) — install to home screen	No store discovery; iOS push limited; APK via F-Droid/sideload

Cost today	₹0 replacement	Trade-off
SMTP	Supabase built-in / Brevo free (300/day) / Resend free (3k/mo)	Rate-limited at scale
Razorpay / FCM / CI / monitoring	Per-txn only · FCM free · public-repo Actions · Uptime Kuma self-host / Sentry free	—

Result: ₹0 recurring cash = Always-Free VM (pipeline) + Cloudflare Pages/R2 (frontend) + Supabase free + templated/free-tier AI.

Making AI ₹0 (pick one)

- Templated analysis (truly ₹0, recommended): generate the report deterministically from metrics you already compute. No tokens, no limits.
- Free-tier LLM (₹0, capped): route AI through Gemini/Groq/Cloudflare Workers AI free quotas, N calls/user/day. Real LLM, rate-limited, quality < Claude.
- Hybrid: templated by default + one free 'AI polish' call/user/day.

₹0-cost revenue (bootstrap still earns)

- Ads — AdSense (PWA/web) / AdMob (APK): zero cost, pure profit.
- Broker affiliate — ₹300–500/demat signup, near-100% margin. Switch on day one.
- Skip credits/Razorpay initially (complexity + per-txn fee) — monetise on ads + affiliate while validating.

Bootstrap unit economics (1M users, ₹0 cash)

Templated AI (₹0 COGS), free infra, no credits/Razorpay — monetised purely on ads + affiliate. All active users are free (no paid tier), so every report is ad-gated.

Monthly P&L;	₹ / month
Banner ads (300k free active × ₹8.7)	**₹26.10 L**
Rewarded ads (11M reports × ₹0.08)	**₹8.80 L**
Interstitial ads	**₹3.00 L**
Broker affiliate	**₹5.00 L**
TOTAL REVENUE	**₹42.90 L**
LLM cost (templated / free-tier AI)	**₹0**
Infra (self-host on Always-Free VM + Cloudflare)	**₹0 cash**
Payment fees (no payments)	**₹0**
NET / month (cash)	**+₹42.90 L**
NET / year (cash)	**+₹5.15 Cr**

vs base case +₹1.76L/mo and after-plan +₹59.4L/mo. Bootstrap nets less than the full paid plan (no credit/Pro revenue, ~99%-margin Claude) but at ₹0 cash — paid instead in ops time, free-tier ceilings (self-host needed past ~thousands of actives) and reliability risk. Rewarded-before-every-report is aggressive UX; trim it and rewarded revenue falls proportionally.

CFO caveats — what ₹0 actually costs

- Founder time (the real bill): self-host + ops ≈ hours/week.
- Reliability/DR: Always-Free reclamation + no backups = you own the risk.
- Ceilings: free tiers cap ~thousands of active users / 500MB before you must pay.
- Brand/trust: *.pages.dev + PWA + templated AI look less premium.
- Still not zero if you earn: GST registration + basic legal are unavoidable (regulatory, not tooling).

Recommendation: bootstrap to product-market fit on ₹0 (Cloudflare Pages + Supabase free + templated AI + ads/affiliate); the moment revenue clears ~₹5k/mo, graduate to branded domain → Supabase Pro → cached Claude. Spend earned money on the premium stack, not pre-revenue cash.

Appendix B — Free / Local AI for Cached Reports (extra cost-down)

Cached reports are generated once per stock per day in the nightly pipeline — a batch, offline job with no per-user latency. That means you don't need a fast, always-on, scalable endpoint; you just grind through N stocks overnight. This is the ideal case for a free or local model, taking the §6 caching lever from ~₹2.4L/mo down to ≈ ₹0 LLM cost.

Option	Cost	Quality	Throughput reality
Local GPU PC you own (Ollama + Qwen2.5-7B / Llama-3.1-8B)	₹0 (electricity)	Good	RTX 3060+ ~30–60 tok/s → ALL 5,800 reports in a few hours. Best true-₹0 path.
Free API tier (Gemini Flash / Groq / Cloudflare Workers AI)	₹0 in limits	Best	~1–2k reports/day free per provider → cover top stocks; spread across providers for more.
Local on Oracle Always-Free ARM VM (CPU, no GPU)	₹0	OK	~8–15 tok/s → ~60–90s/report → only ~200–300 stocks/night before saturation.

Throughput math & the hybrid pattern

- 5,800 stocks on the free ARM CPU ≈ 120+ hours (won't finish overnight). Top ~250 ≈ 4–6 hrs (fits). GPU / free API → full universe nightly.
- Recommended hybrid: generate the top ~200–500 stocks with a real (local/free) model; template the long tail from metrics + patterns.ts (already ₹0). Users mostly view the popular names.

Commercially-free models & guardrails

- Free for commercial use: Qwen2.5 / Mistral (Apache-2.0, safest), Llama 3.x (free under 700M MAU), Gemma 2 (Gemma terms). Run via Ollama or llama.cpp.
- Constrain to summarise facts, not invent them — feed the precomputed metrics/patterns and forbid new numbers (the ai-report function already does this). Small models hallucinate more, so this matters.
- Keep the templated fallback + SEBI disclaimer — factual, non-advisory, no targets/recommendations.

Pipeline fit

Add a nightly step in backend/app/dataengine: after export → for the top-N symbols call local Ollama (or a free API) → write reports/<SYMBOL>.json → ship in the bundle / push to Cloudflare R2 → served from CDN at ₹0 marginal cost. The existing ai-report Edge Function becomes the on-demand fallback for non-cached stocks. Net effect: the LLM line collapses to near ₹0 at any scale.

Prepared as an internal CFO review. Figures are indicative planning estimates at conservative assumptions; verify current vendor pricing and validate conversion/eCPM with live data before committing spend.